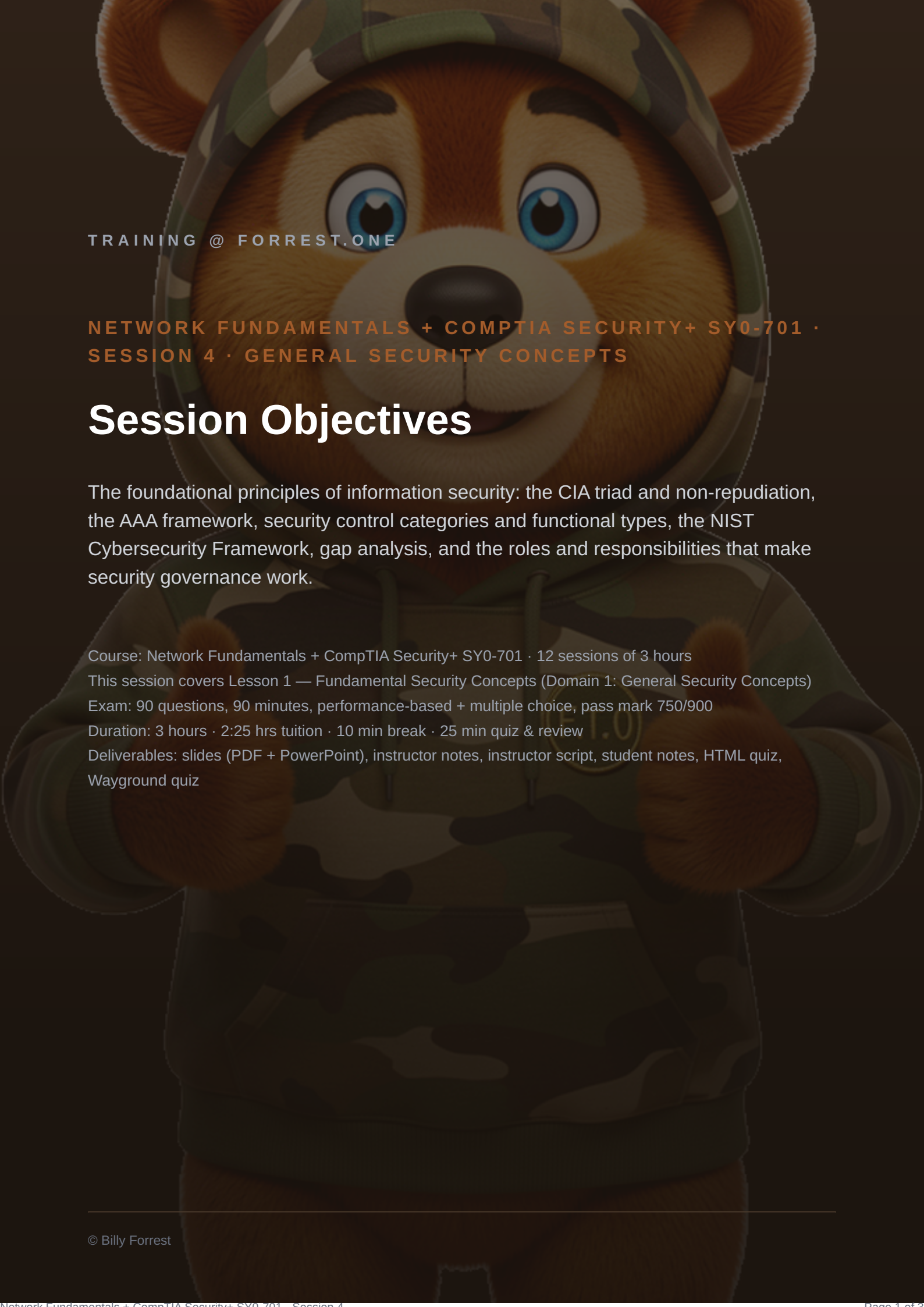




TRAINING @ FORREST.ONE

NETWORK FUNDAMENTALS + COMPTIA SECURITY+ SY0-701 ·
SESSION 4 · GENERAL SECURITY CONCEPTS

Session Objectives

The foundational principles of information security: the CIA triad and non-repudiation, the AAA framework, security control categories and functional types, the NIST Cybersecurity Framework, gap analysis, and the roles and responsibilities that make security governance work.

Course: Network Fundamentals + CompTIA Security+ SY0-701 · 12 sessions of 3 hours

This session covers Lesson 1 — Fundamental Security Concepts (Domain 1: General Security Concepts)

Exam: 90 questions, 90 minutes, performance-based + multiple choice, pass mark 750/900

Duration: 3 hours · 2:25 hrs tuition · 10 min break · 25 min quiz & review

Deliverables: slides (PDF + PowerPoint), instructor notes, instructor script, student notes, HTML quiz, Wayground quiz

Session Objectives — Fundamental Security Concepts

This first session of the Network Fundamentals + CompTIA Security+ SY0-701 course establishes the foundational principles of information security, matching Lesson 1 of the Cyber Academy programme and Domain 1 of the exam (General Security Concepts, 12% of the paper). It requires no prior security knowledge: everything builds from the CIA triad and from the controls that protect it.

Learning outcomes

By the end of the 3-hour session, candidates will be able to:

- Explain why security controls exist and how layered defence (defence in depth) reduces risk.
- Classify security controls by category: technical, physical, managerial and operational.
- Classify security controls by functional type: preventive, detective, corrective, compensating, deterrent and directive.
- Define confidentiality, integrity and availability, and give a real-world example control for each.
- Explain non-repudiation and how digital signatures and audit logs provide it.
- Describe the three authentication factors and the role of Kerberos, LDAP, SAML and SSO.
- Explain accounting and audit trails, and the three duties of log management (centralise, retain, analyse).
- Describe zero trust: never trust always verify, microsegmentation, continuous verification and least privilege.
- Explain deception technologies: honeypots, honeynets and honeytokens, and what they buy you.
- Walk the four steps of a controlled change and the four technical implications of change.
- Describe the NIST Cybersecurity Framework functions and explain what a gap analysis measures.
- Name the key security roles and their responsibilities: CISO/CSO, ISSO, data owner, data custodian, analyst, auditor.

CompTIA exam context

Item	Detail
Exam	Network Fundamentals + CompTIA Security+ SY0-701
Questions	90 (multiple choice + performance-based)
Duration	90 minutes
Pass mark	750 on a scale of 100–900

Item	Detail
Domain 1 weight	General Security Concepts, 12%
Vendor	Vendor-neutral, no product-specific knowledge required

Session coverage (aligned to the expanded narration)

Chapter	Title	What it covers
1	Why Controls Exist	Controls as safeguards; external and internal threats; the layered view and defence in depth
2	Classifying Controls by Category	Technical, physical, managerial and operational controls
3	What Each Control Type Does	Preventive, detective, corrective, compensating, deterrent and directive
4	Proving Identity and Keeping Records	CIA in practice, non-repudiation, authentication factors and protocols
5	Accounting and Zero Trust	Audit trails, log management, never trust always verify, continuous verification
6	Deception and Change Control	Honeypots, honeynets, honeytokens, change management and its technical impact
Exam	Domain 1 alignment	NIST CSF, gap analysis, core principles, roles and responsibilities

How this session maps to the exam

Every activity and quiz question is tagged with the exam objective it tests (e.g. 1.2 Security Controls). Using the session quiz to 70%+ before the exam is the fastest route to Domain 1 marks. The two-axis control model (category and functional type) appears in more Domain 1 questions than any other single idea.